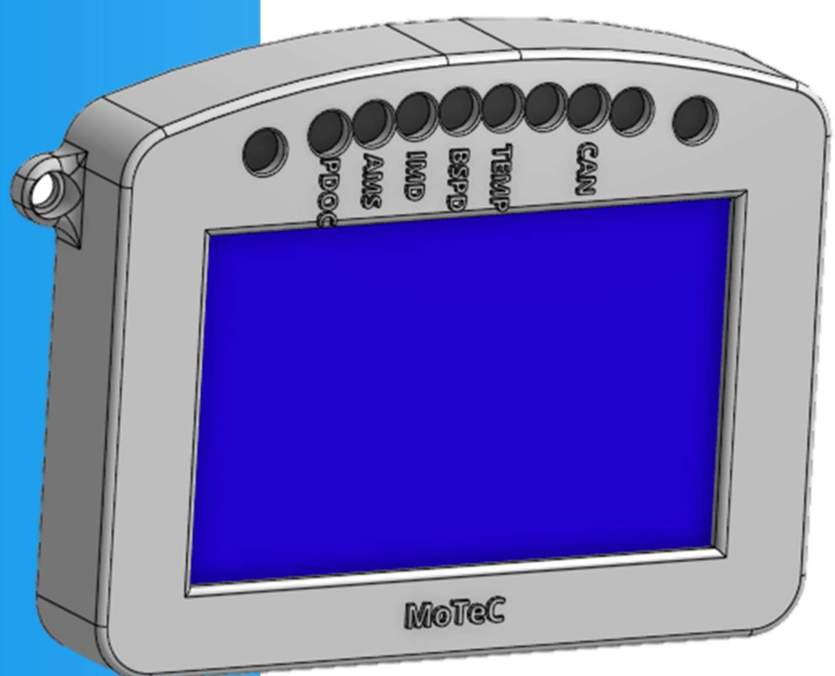
MoTeC Zero-To-Hero

ZERO-TO-HERO



Contributors:

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Jayson Mills

2025

Introduction – PLEASE READ

The information contained within this document is a combination of knowledge over the last 3 or 4 years. From the author’s knowledge, this information has not been recorded in any NU Racing reports.

This Zero-To-Hero has been made on a Windows 11 machine. The information should carry to Mac and Linux machines, if Dash Manager, Display Creator, T2 and i2 all have installers for Mac and Linux. The sections requiring folder searching will be the main difference.

As of writing, this Zero-To-Hero is saved to the NU Teams Sharepoint, the author hopes that this Sharepoint still exists. Once this Zero-To-Hero is finished (as finished as the author can get it) it will also be saved to the ‘Racing-MoTeC’ repository. If edits are made to this file, care should be taken to ensure that these versions remain the same, to avoid new information not being followed correctly.

Contents

Racing-MoTeC Repository

The place for all Racing MoTeC related information. Without this repo installed on the computer that's being used, none of the steps given here can be done

Installing Racing-MoTeC Repository

- Clone from Github (ask Mechatronic Department Leader or Chief Engineer for access)
- Usually, it is encouraged to make a separate folder for all Github repos, that is outside of the OneDrive folder
- Once successfully cloned, the below steps can be followed

Setting up MoTeC for a Track Day

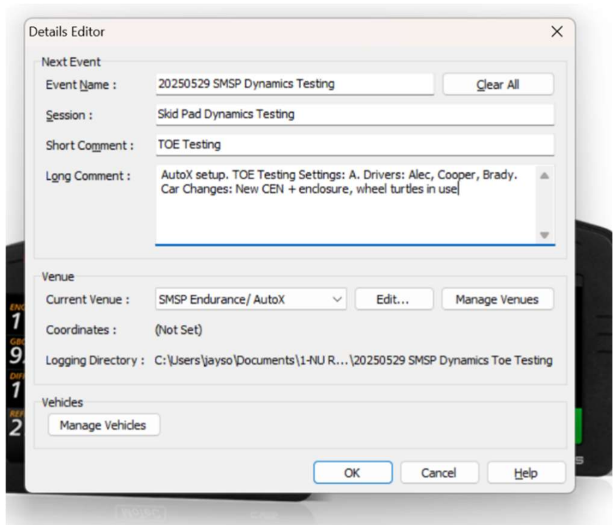
Track Day Setup Checklist

☐ Pack MoTeC at TA
☐ Set up laptop and clear data from the car
☐ Connect the transceiver box and test live data
☐ Power the lap beacon
☐

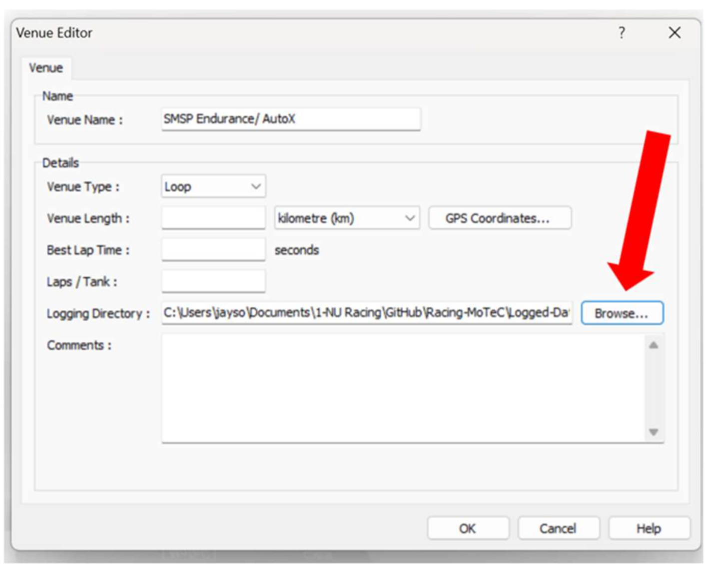
Clearing Data Logger at the start of the Day

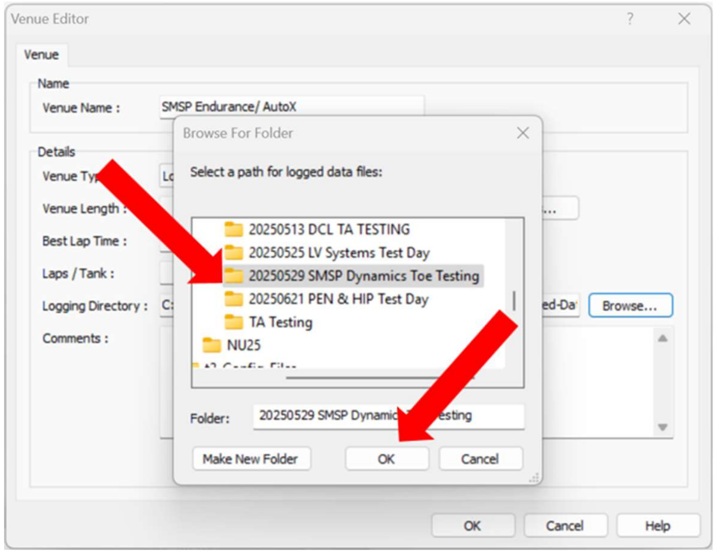
Prior to running the car, the logging memory within the dash must be cleared to eliminate any excessive power cycles or TA testing being included within the track day logging files. To do this the data must be pulled (see Data Manager- Pulling Data) and the option for clearing logging memory selected. The directory location of this initial pull may depend on the extent of the prior data. If extensive relevant data is present, it may be preferable to save this data in a separate TA testing folder under with the Racing MoTeC GitHub repo 'Racing-MoTec > Logged-Data > NUXX'. Otherwise, this data may be saved under the track day folder with the session comment "Initial Clear". For each track day or testing event, create a new folder within 'Racing-MoTec > Logged-Data > NUXX' with the convention of YEARMONTHDATE "Track Day Title" (e.g. "20250524 Uon LV Systems Track Day").

The initial clear is a good opportunity to set up the **event details**, venue and vehicle for the entire day. Every time you get and clear logged data you set the event details of that individual pull but having most of this prefilled will save time. The event details should clearly define the testing that is commencing, any changes and issues with the car, who are the drivers and anything else important to note when looking at the data. An example of filling out these fields is demonstrated below.



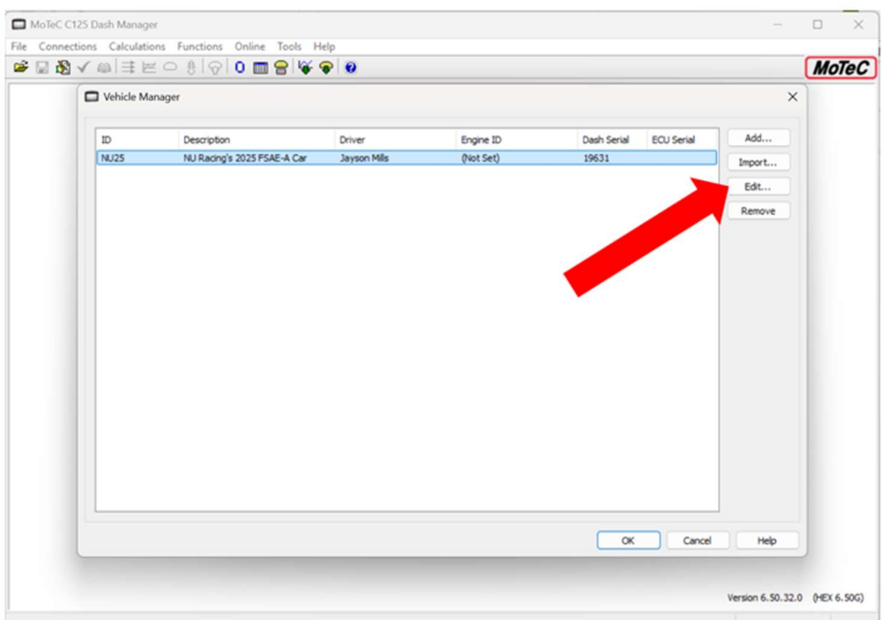
The venue must also be set as shown above. Most importantly the logging directory MUST be selected (sets the folder where the files save). Do this by selecting Edit… in the venue area where the screen below will appear. Hit Browse… in the logging directory field and select the desired folder. Click OK to confirm your folder selection.



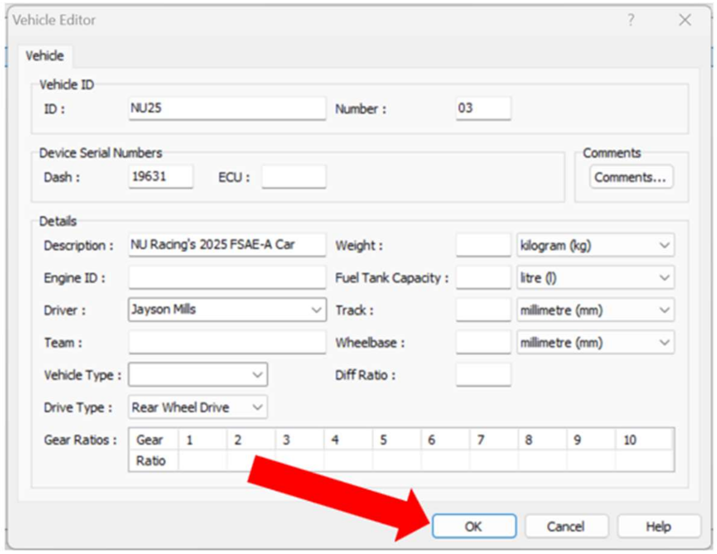


You may now click OK on the venue editor. Double and Triple check this logging directory is properly selected by re-entering the venue editor and checking the file path. You do not want to be saving data throughout the day into a random folder… embarrassing.

In the vehicle section select Manage Vehicles and the menu depicted below will appear. From here a vehicle can be added, imported, edited or removed. If the car you are trying to select it not present, press Add… to create a new vehicle. Otherwise use Edit… to make any changes.



In the edit screen, the car's details may be filled in. Most important fields are the Vehicle ID, the dash serial and driver names. It should be noted that for each data pull, the driver should be set to the first driver and new drivers are added using this menu. This menu is shown below. Hit OK when finished.



With the data log cleared and the event details filled out, the reader may then move on to other steps. It should be reiterated that for each data pull the event details are set for the data logged before that pull. Use this to your advantage by changing any event fields that are useful in discriminating between data files. For example add comments such as “BMS fault P0A07 during 2nd enduro run”, “ CEN sounder not operational”, “First Driver: Alec (Regen on)” etc.

Transceiver Box

T2

i2

Dash Manager

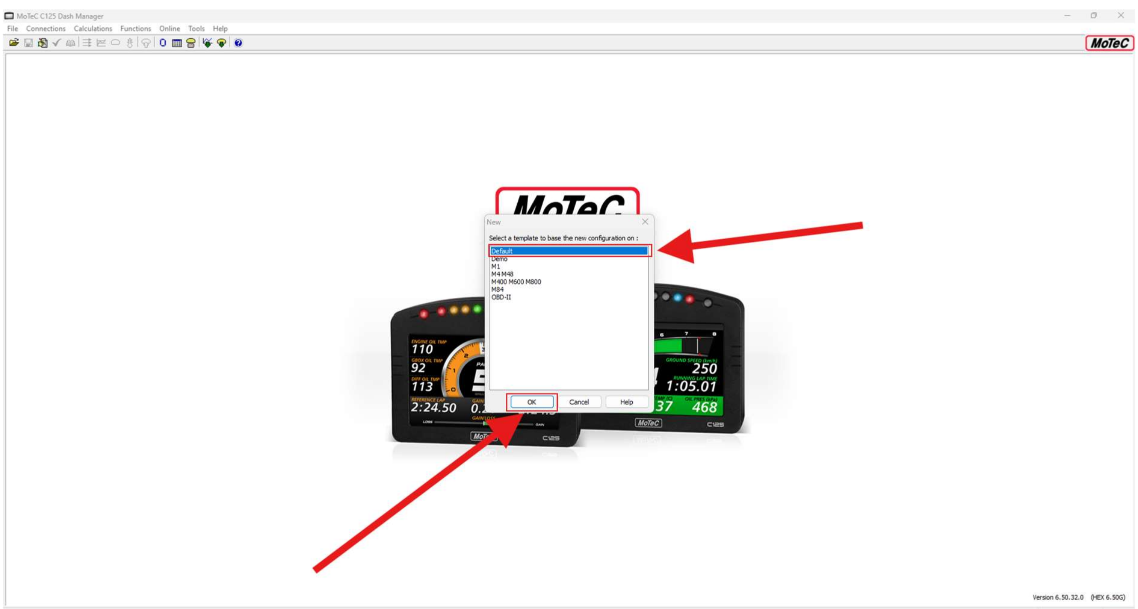
Dash manager is used for all communication with the C125 MoTeC Display

Configuring Dash Manager (First Time Set-up)

Creating a Configuration File

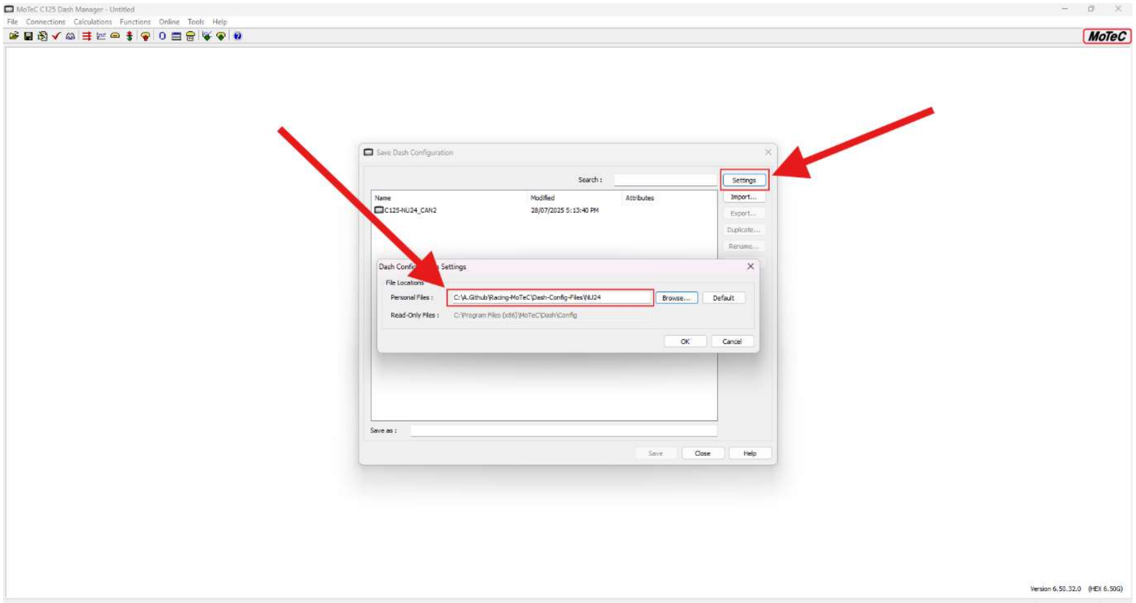
- Select File in the top right-hand corner
    - From the dropdown, select 'New'

- Select the desired template, usually Default is fine, then select 'Ok'

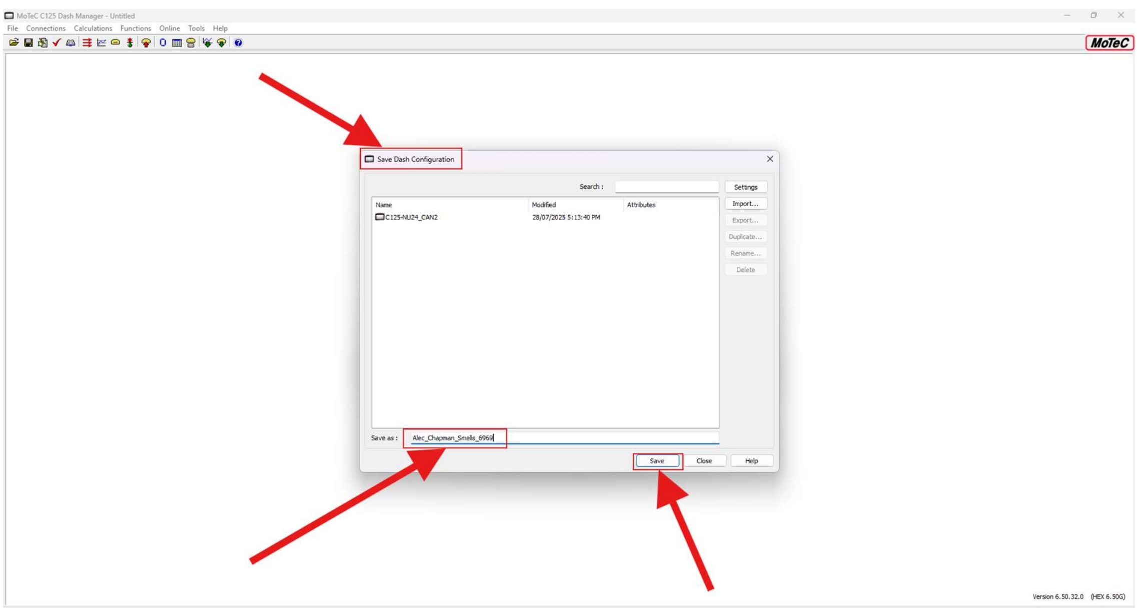


- The new file will be loaded
- To save changes and give the new config a name, select 'File' in the top right-hand corner again
    - From the dropdown, select 'Save As'. A menu titled 'Save Dash Configuration' should appear
- To ensure that the file path for the new config file is correct, select 'Settings'
    - From there, ensure that the file path is set to the 'Dash-Config-Files\'car_name'

- TODO: the author wants to change this so that it isn't a dash config per car, instead be one main config file that all the NU Racing electric cars use. This requires one main DBC file that doesn't get a new iteration each year (etc, EV3, NU23, NU24, …) which the author also hasn't done yet



- Give the config file a name, then click 'Save'
  - This should save it to the correct folder. Double check by going into the folder and checking that it has been saved there

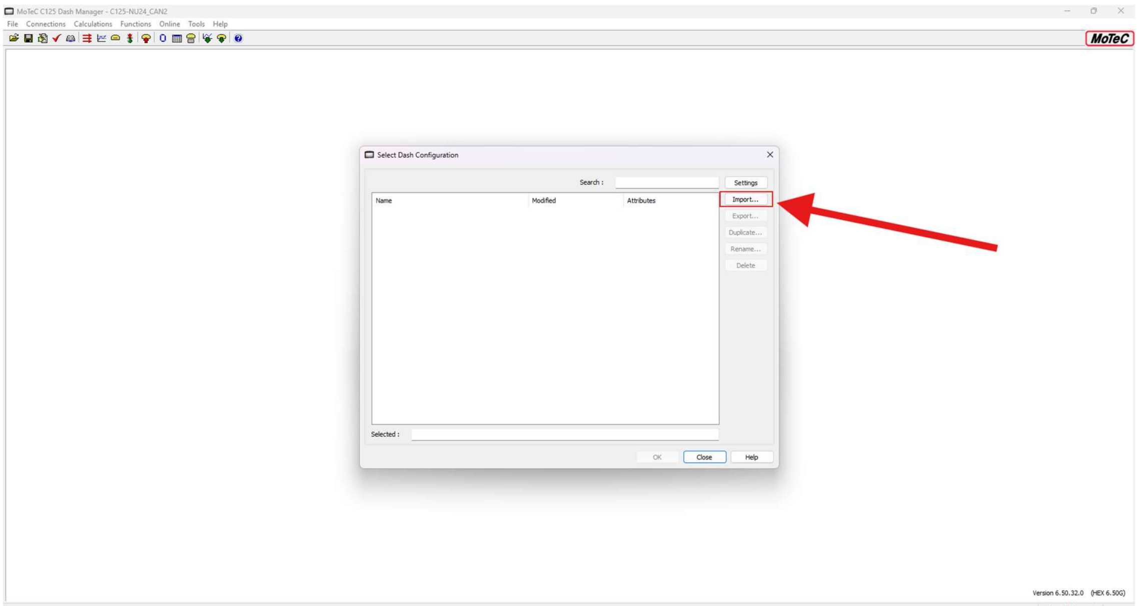


- Then make sure to click 'Save File' in the top left-hand corner
    - Dash Manager is a funky program, and if this isn't done then changes won't update through the whole program correctly

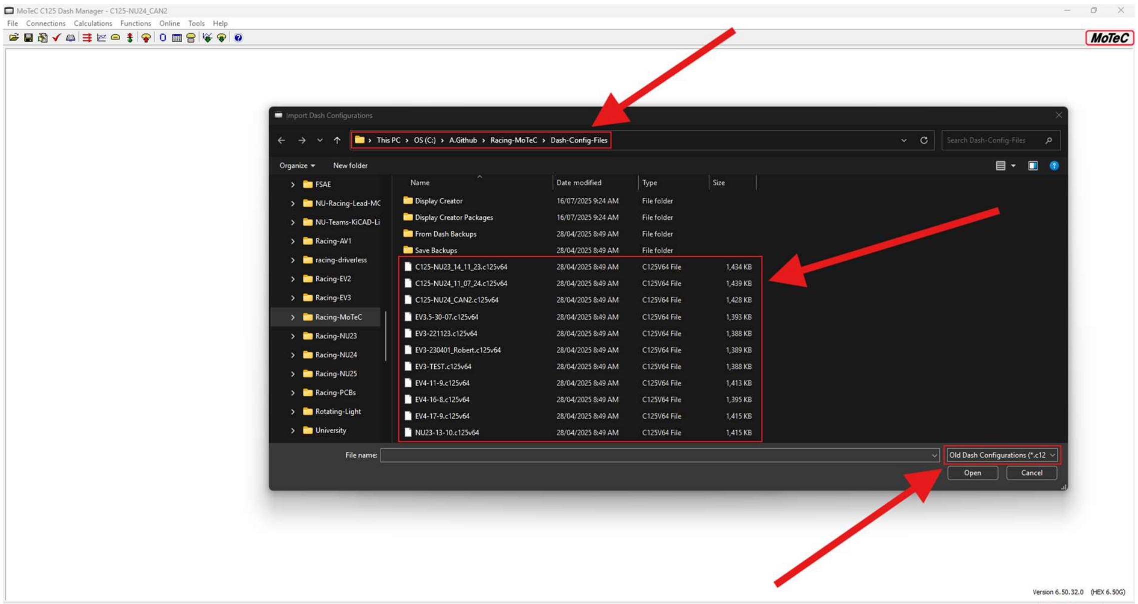
Opening an Existing Configuration File

- Open MoTeC C125 Dash Manager on a laptop/PC
- Select 'Open File'

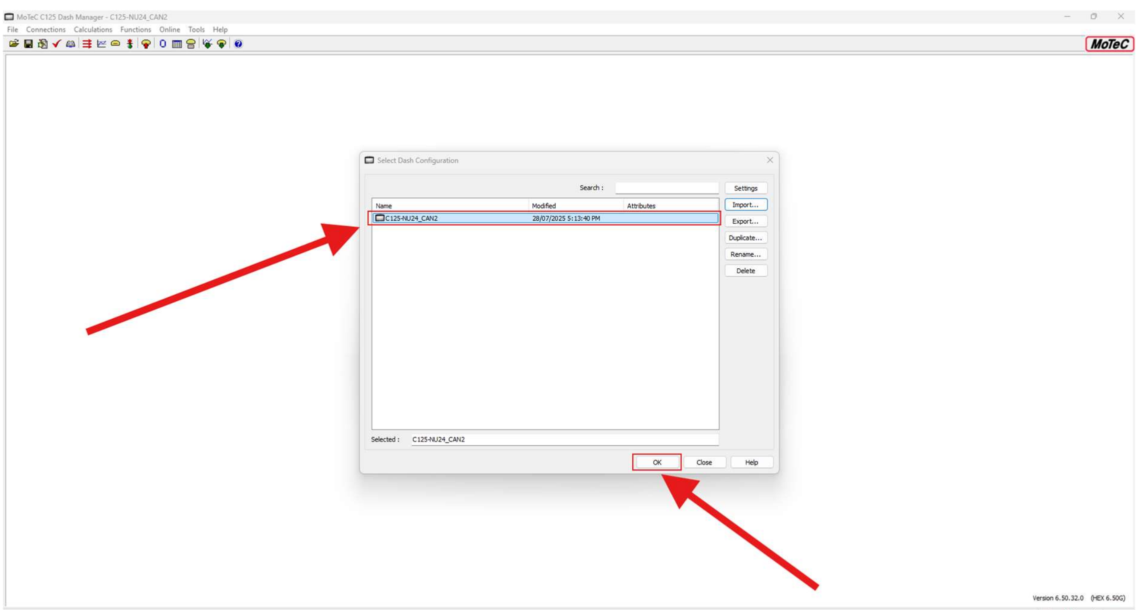
- Select 'Import…'



- Navigate to ‘Racing-MoTeC > Dash-Config-Files’
    - See the Racing-MoTeC Repository section if this repo is not yet installed from Github
    - If there are no files present, the current config file has been made with an older version of the config file type (pre v65), to fix this, ensure that the file type is set to ‘Old Dash Configurations (*.c125v15…*.c125v64)’ in the lower right-hand corner
    - Once the files are visible, select the desired config file. For this demonstration, the ‘C125_NU24_CAN2.c125v64’ will be used



- Once desired config file is selected and imported, it should appear as shown below, then simply select it and click ‘Ok’, it will now be loaded into Dash Manager



- Then make sure to click 'Save File' in the top left-hand corner
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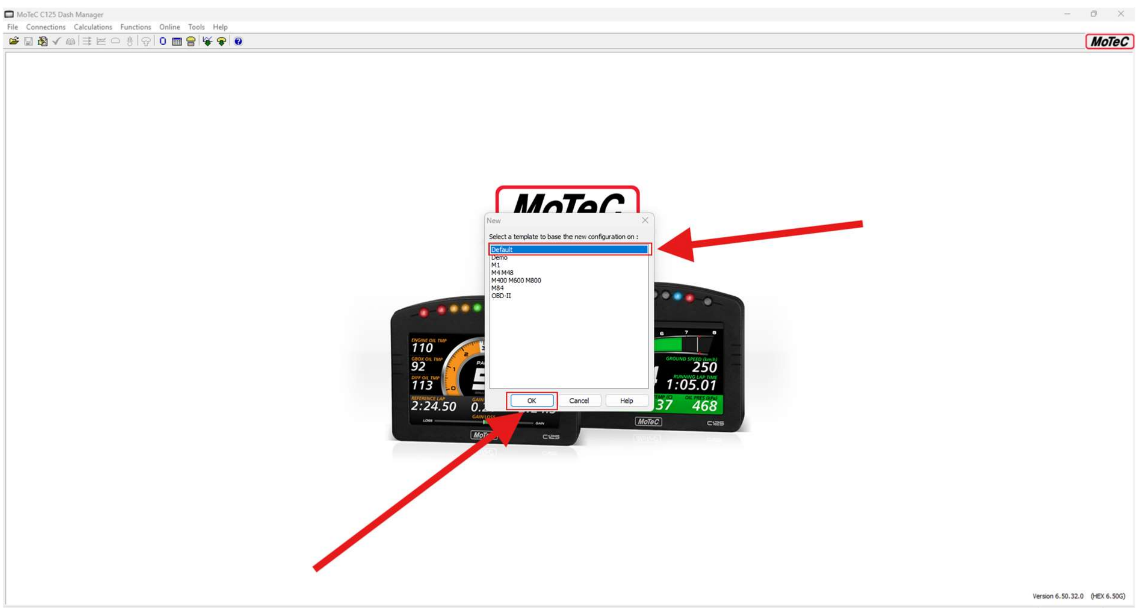
Adding Messages

After opening a configuration file (see the Creating a Configuration File

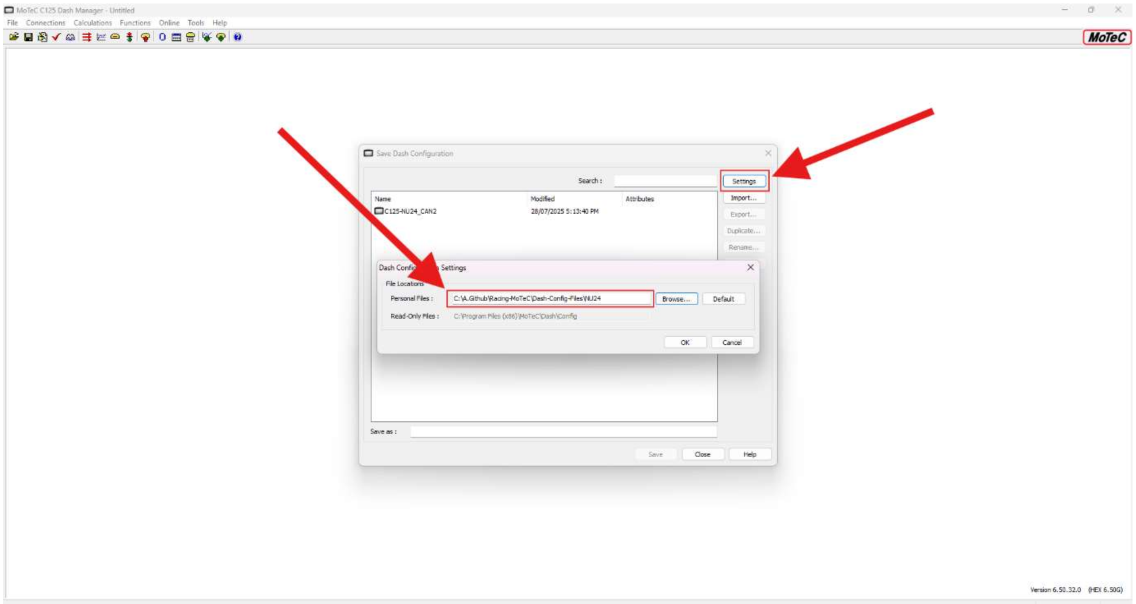
- Select File in the top right-hand corner
  - From the dropdown, select 'New'

- Select the desired template, usually Default is fine, then select 'Ok'

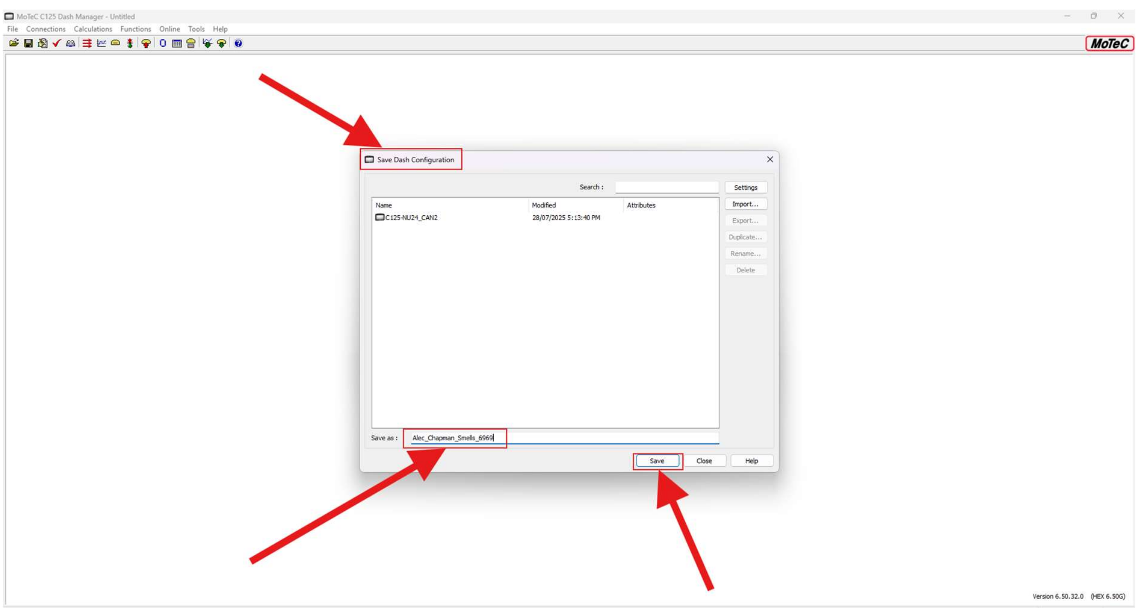


- The new file will be loaded
- To save changes and give the new config a name, select 'File' in the top right-hand corner again
  - From the dropdown, select 'Save As'. A menu titled 'Save Dash Configuration' should appear
- To ensure that the file path for the new config file is correct, select 'Settings'

- From there, ensure that the file path is set to the 'Dash-Config-Files\'car_name'
  - TODO: the author wants to change this so that it isn't a dash config per car, instead be one main config file that all the NU Racing electric cars use. This requires one main DBC file that doesn't get a new iteration each year (etc, EV3, NU23, NU24, …) which the author also hasn't done yet



- Give the config file a name, then click 'Save'
  - This should save it to the correct folder. Double check by going into the folder and checking that it has been saved there



- Then make sure to click ‘Save File’ in the top left-hand corner
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- Opening an Existing Configuration File section for more information), the following sections can be followed

To a new Config File…

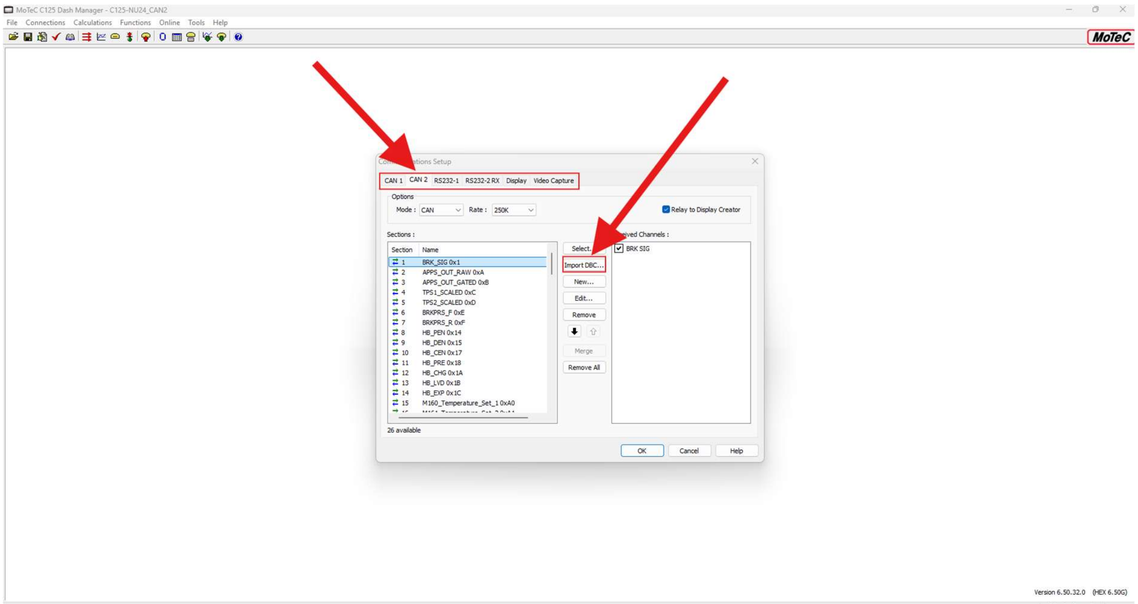
- Follow the same steps as To an existing Config File…, but select all the messages in the DBC that is wanting to be used.

To an existing Config File…

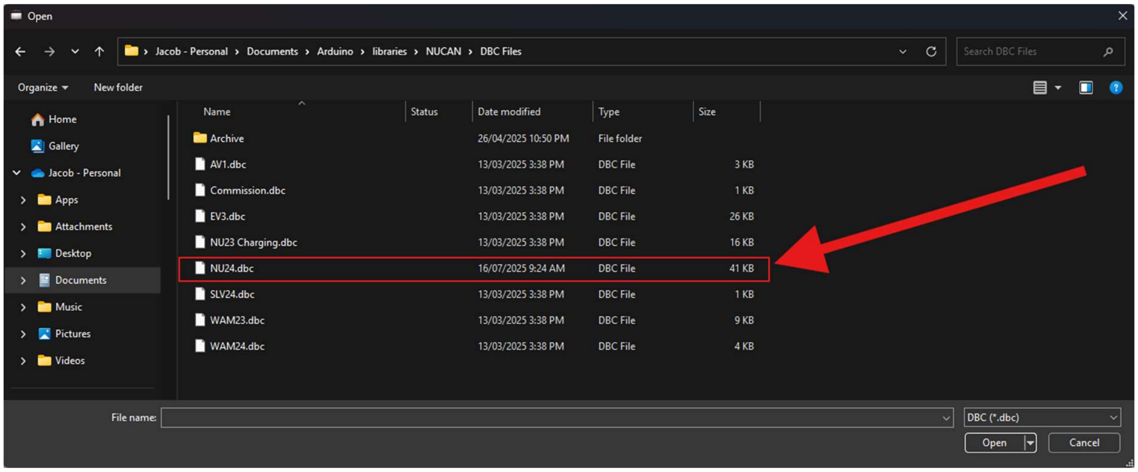
Communications Setup

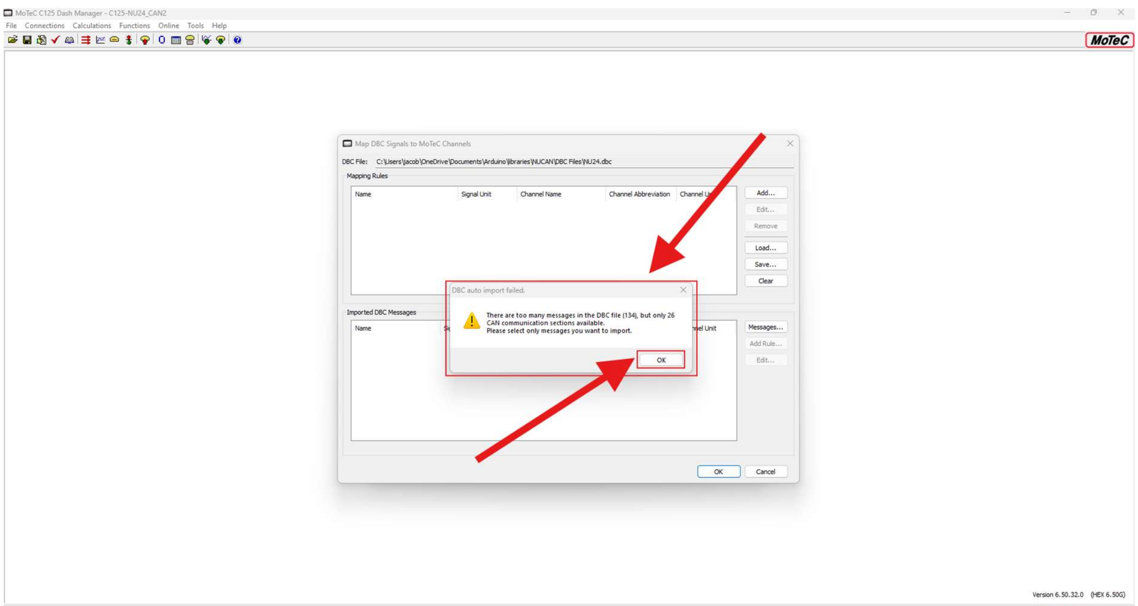
- Select 'Connections'

- From the drop down, click ‘Communications…’
- Select the desired communication protocol
    - At the date of writing, NU Racing uses CAN 2 on the C125 due to issues occurring with CAN 1 in 2024, which has now been fixed, just haven’t moved back to CAN 1 (as of July 2025)
- Import DBC

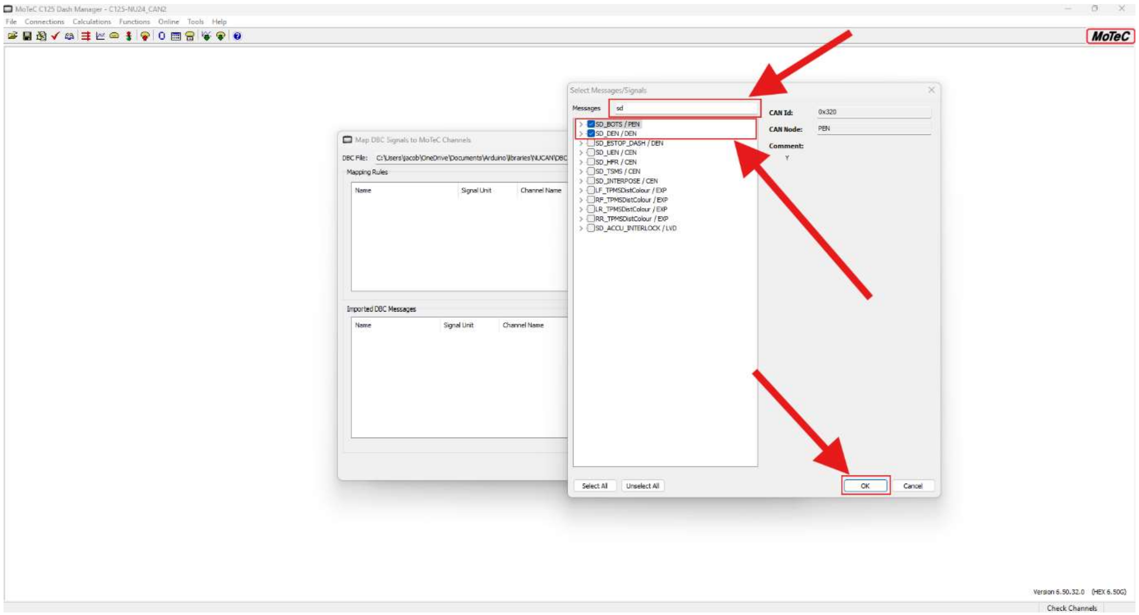


- Navigate to the desired DBC with the new message(s) (e.g. NU24)
    - There should be an error saying that there are too many messages (if adding a newly added message to the current DBC), just click Ok

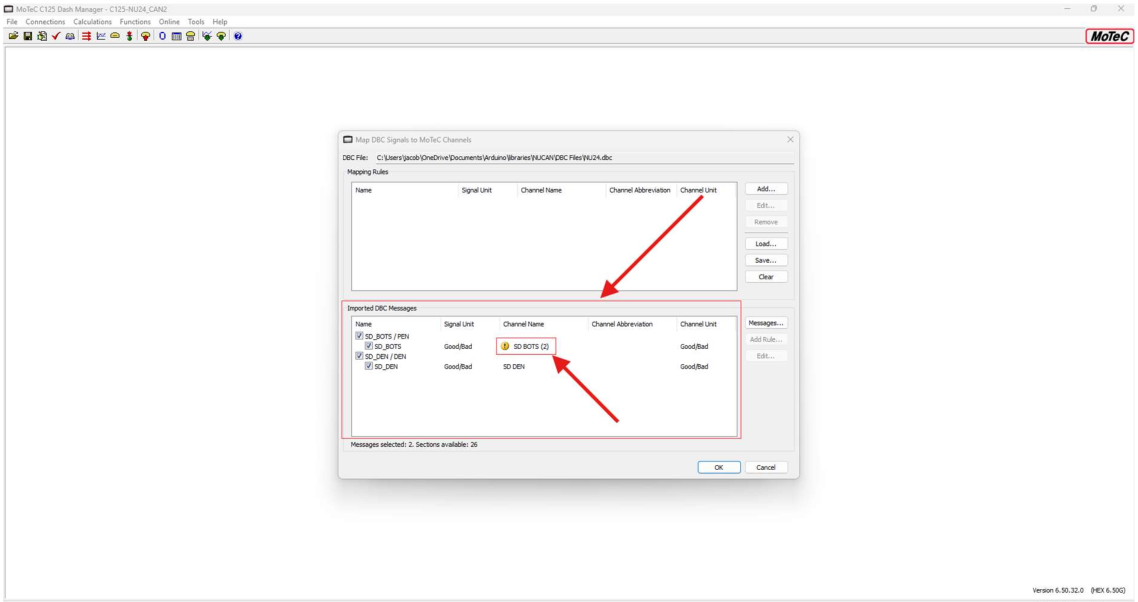


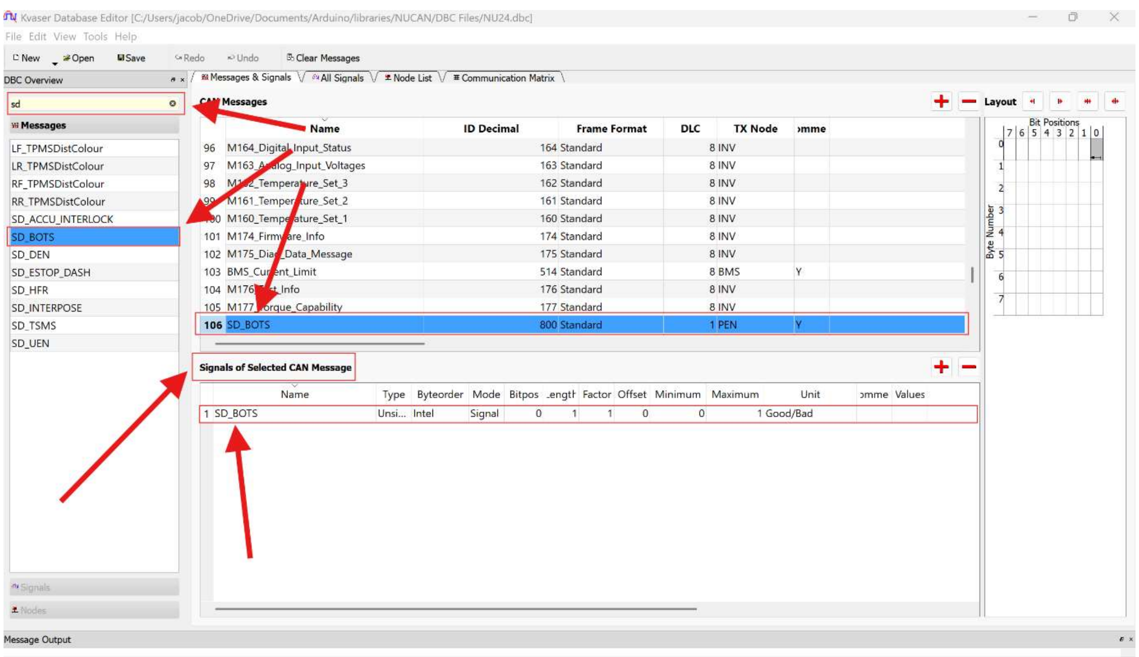


- Select the designed message(s) to import
    - Search button up the top can be used to quickly find the new signal
    - Just a message or multiple messages can be added at the same time
    - Once the desired message(s) are selected, click ‘Ok’

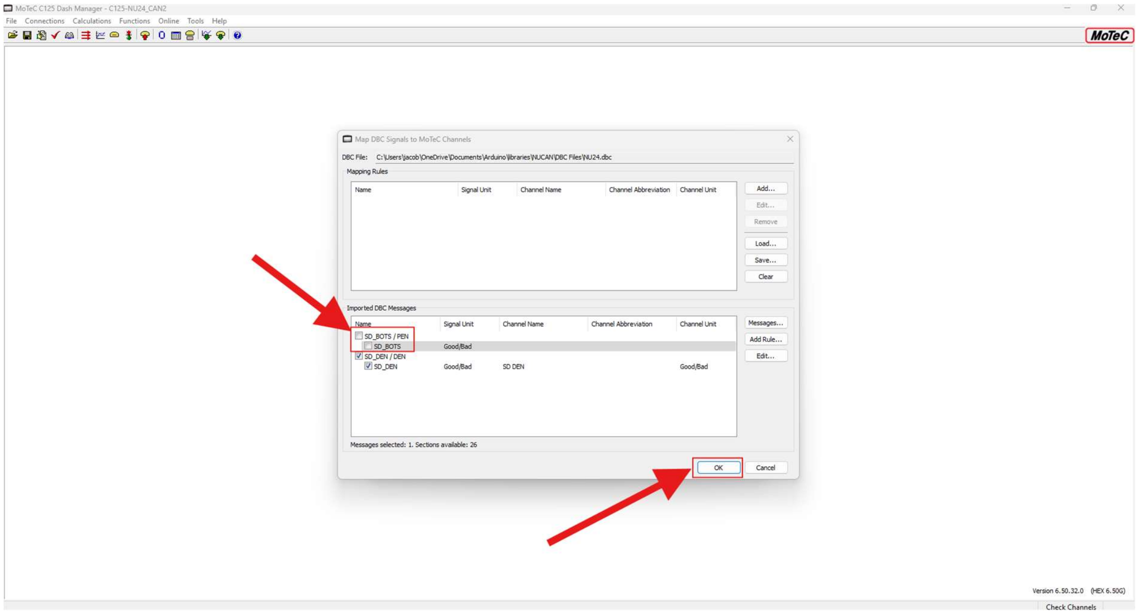


- The message(s) will then be loaded into the ‘Imported DBC Messages’ box
    - If there is a double up, then the ‘Channel Name’ will have a yellow circle with an exclamation mark in the middle. If this is the case, uncheck the box next to either the signal or the message to not import the duplicated message(s). If this is done then there will be message conflicts within the MoTeC (which is not a fun thing to fix)
    - In Dash Manager, Signals are called Channels
    - If this occurs when it shouldn’t, check the signal names of the message inside the DBC, as the Signal name is what is being duplicated and having issues, not the name of the message.
    - The this can be checked and edited within the DBC. Open the DBC in a text file format (if you hate yourself), or through Kvaser Database Editor. See Alec Chapman’s ‘NUCAN ZERO TO HERO’ (hopefully located either in the Sharepoint, or in the NUCAN Repository) for how to read and open a DBC using Kvaser Database Editor
        - Download link for Kvaser Database Editor (chose not to include a link to a specific version as at the time of writing, the most recent version is V3. So just search for Kvaser Database Editor at the following link): https://kvaser.com/download/

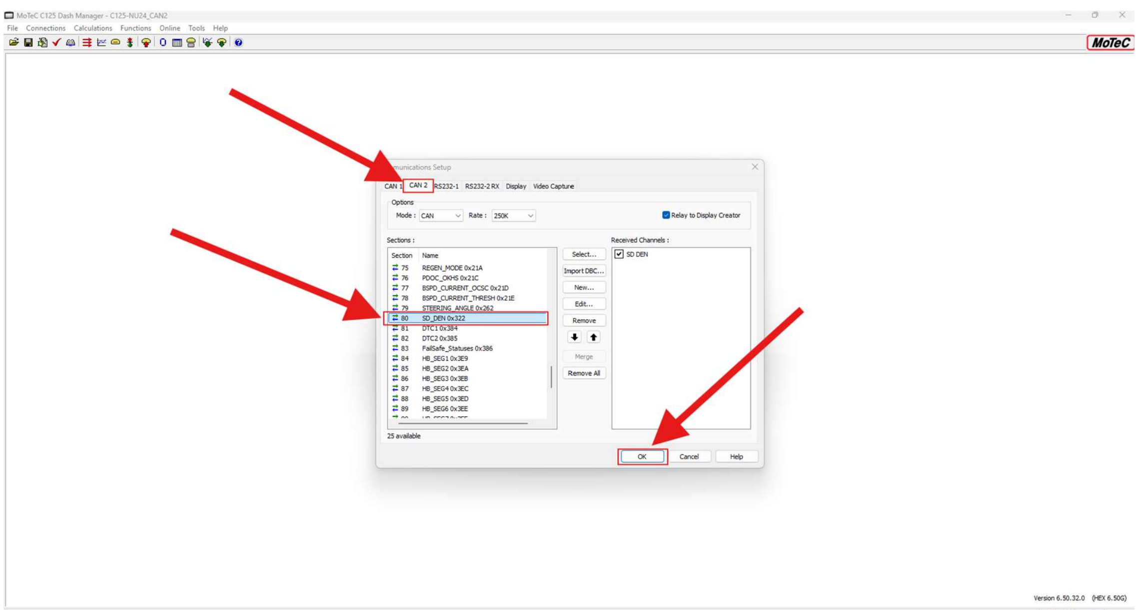




- Once the desired message(s) are selected, click ‘Ok’



- They should now show up in the Communications Setup page under the desired protocol
- Now click Ok again



- Then make sure to click 'Save File' in the top left-hand corner
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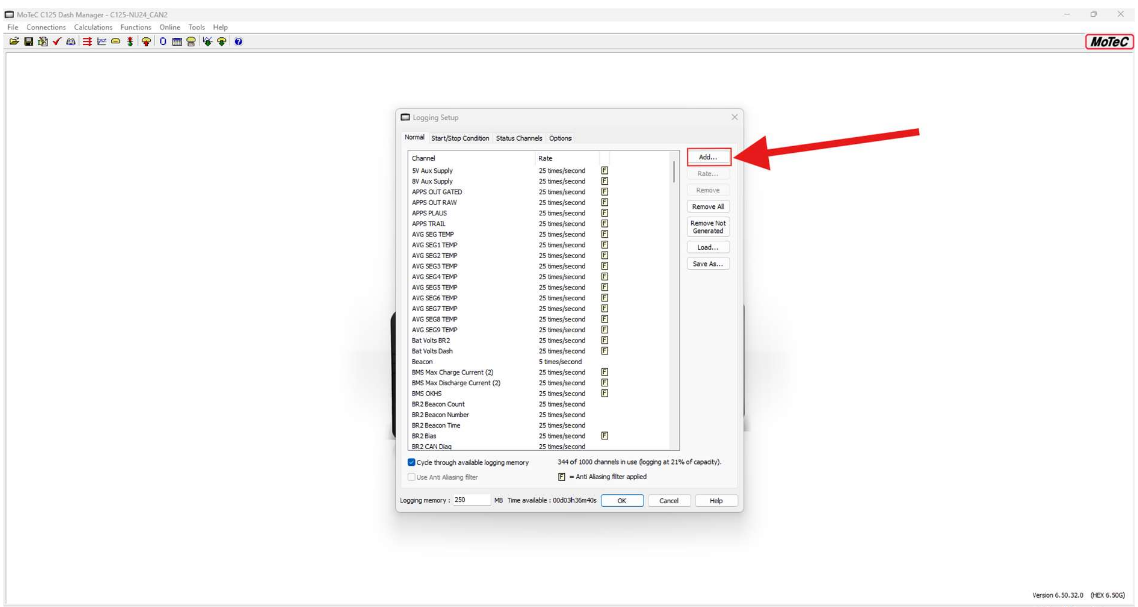
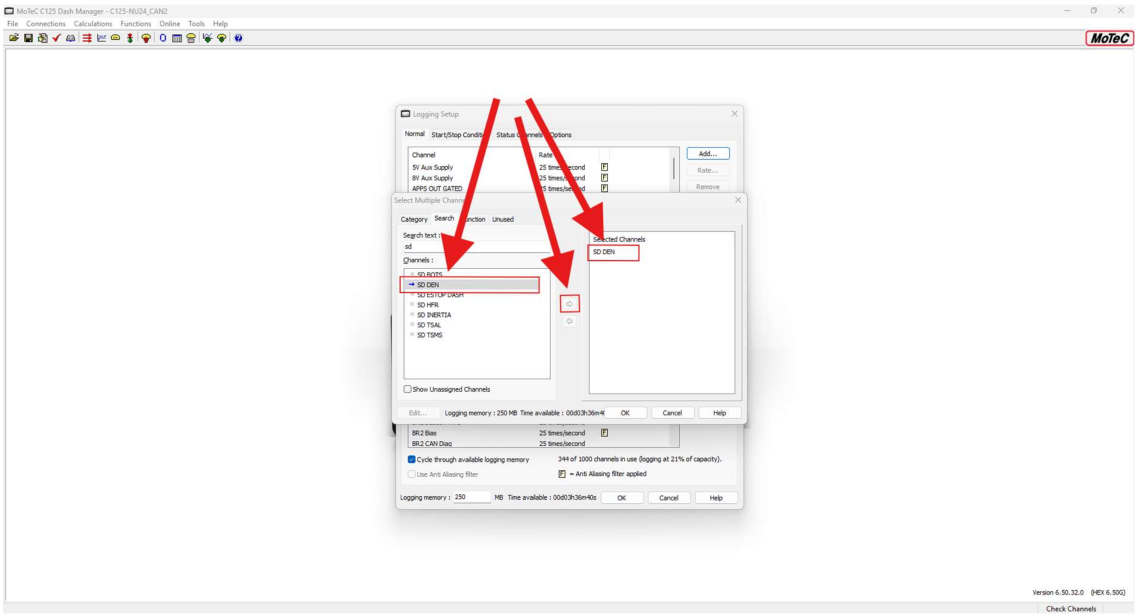
Logging Setup

- Setup Logging

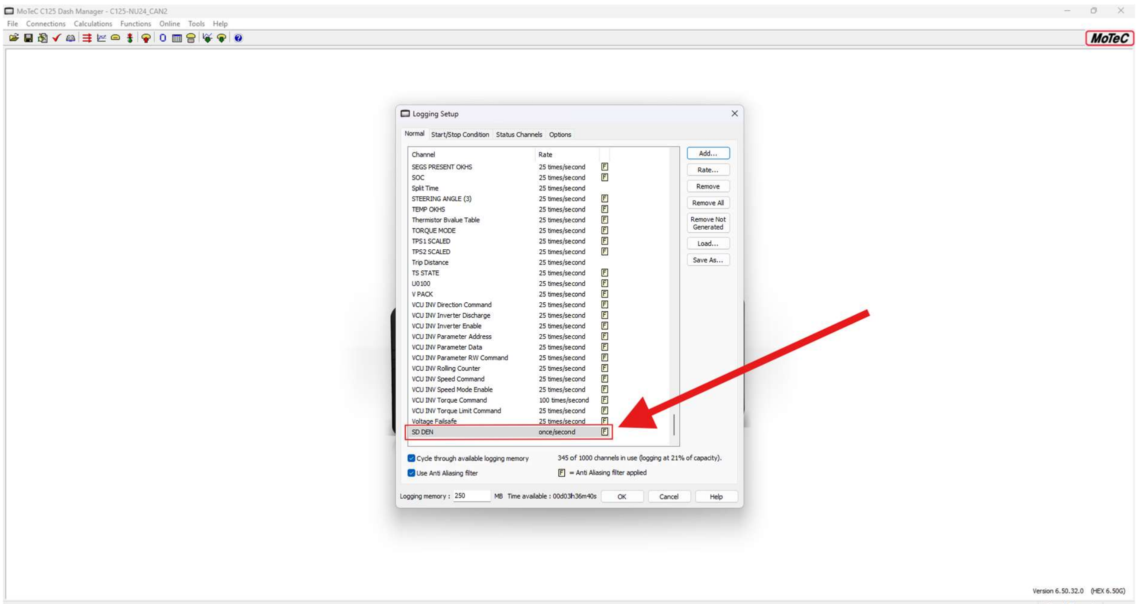
- Add (top right)

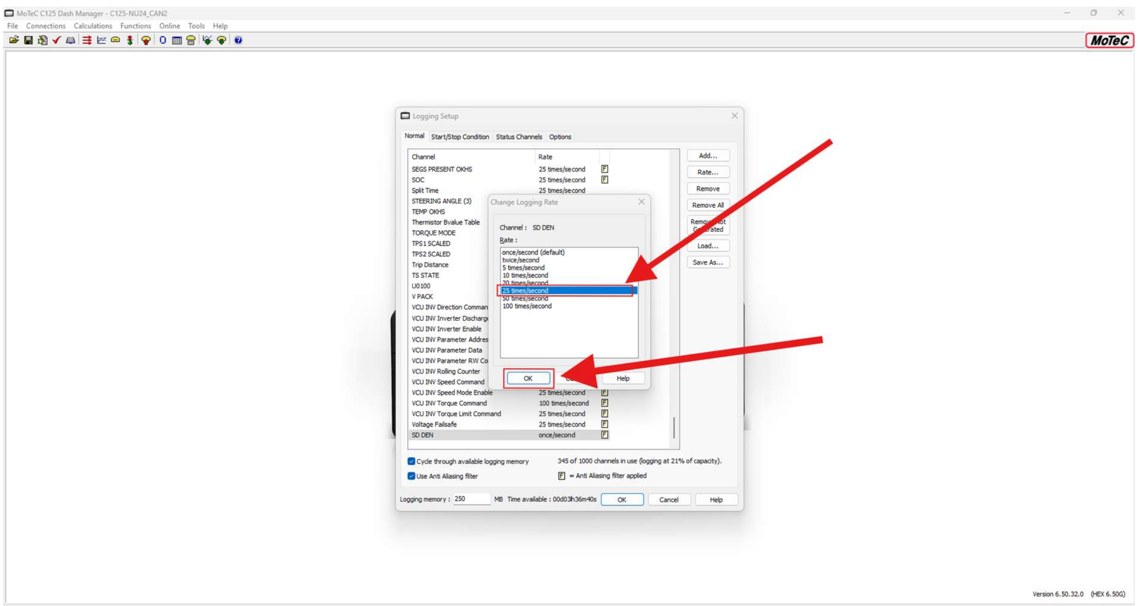


- Find the message(s) that was just added to Communications Setup
  - New messages will have a blue arrow to the left of them, otherwise just search for the new message
- Click the arrow point to the right in the middle of the window
  - The message(s) should now be in the 'Selected Channels' box
- Click Ok to close out of that window

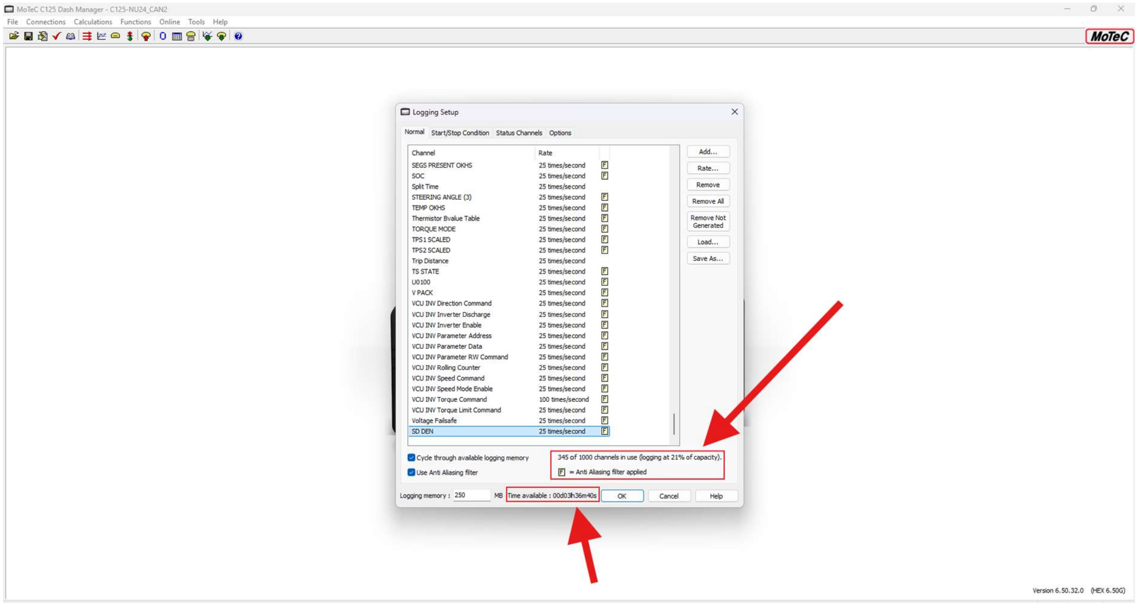


- Scroll to find the new added message(s)
    - Either double click the message(s) or select 'Rate…' at the top right
    - Change the rate to the desired value
        - For most messages, a rate of 25 times/second is quick enough for logging purposes. Some OEM components may send these messages quicker, or require them to be sent quicker, for example, the VCU INV Torque Command shown below is set to 100 times/second. If unsure, consult Mechatronic Department Lead or Chief Engineer
    - Select 'Ok'





- Take note of the number of channels used, capacity, and 'Time available' shown at the bottom. If channels used or capacity are nearing their limit take note and inform the Mechatronic Department Leader or Chief Engineer
    - If Time available is nearing to zero, inform the Mechatronic Department Leader or Chief Engineer

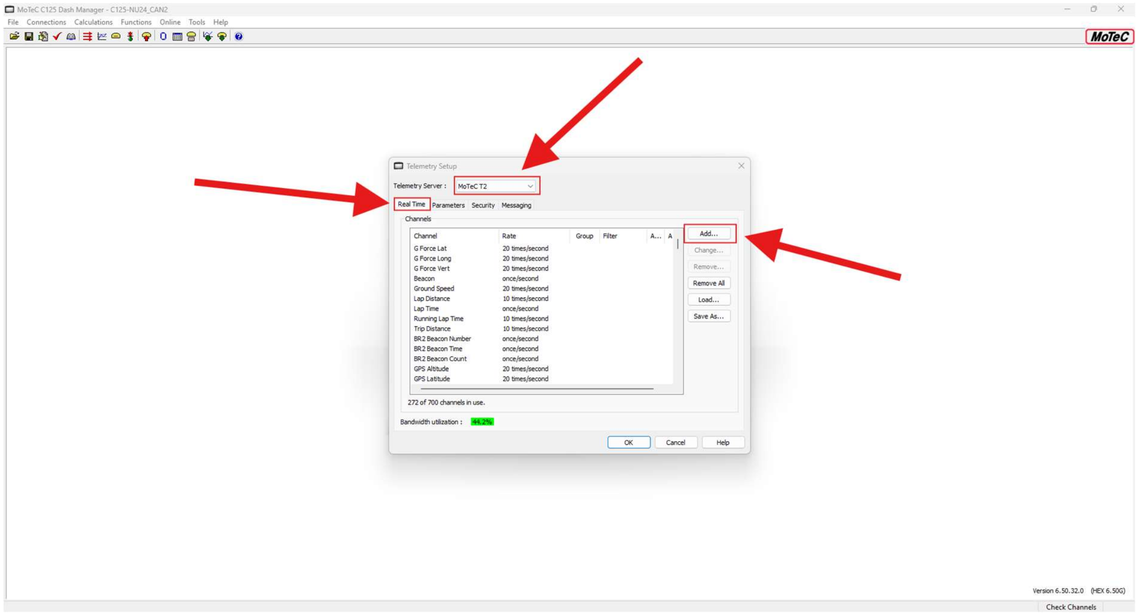


- Click 'Ok' and to leave the Logging Setup
- Then make sure to click 'Save File' in the top left-hand corner
    - Dash Manager is a funky program, and if this isn't done then changes won't update through the whole program correctly

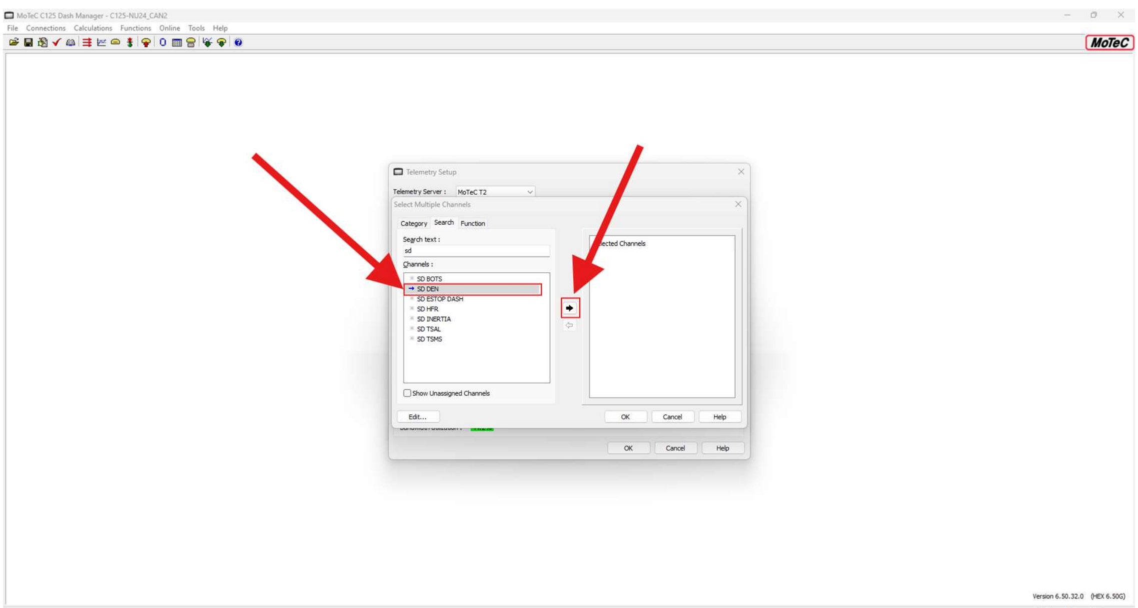
Telemetry Setup

- Select ‘Functions’, then from the dropdown, select ‘Telemetry…’

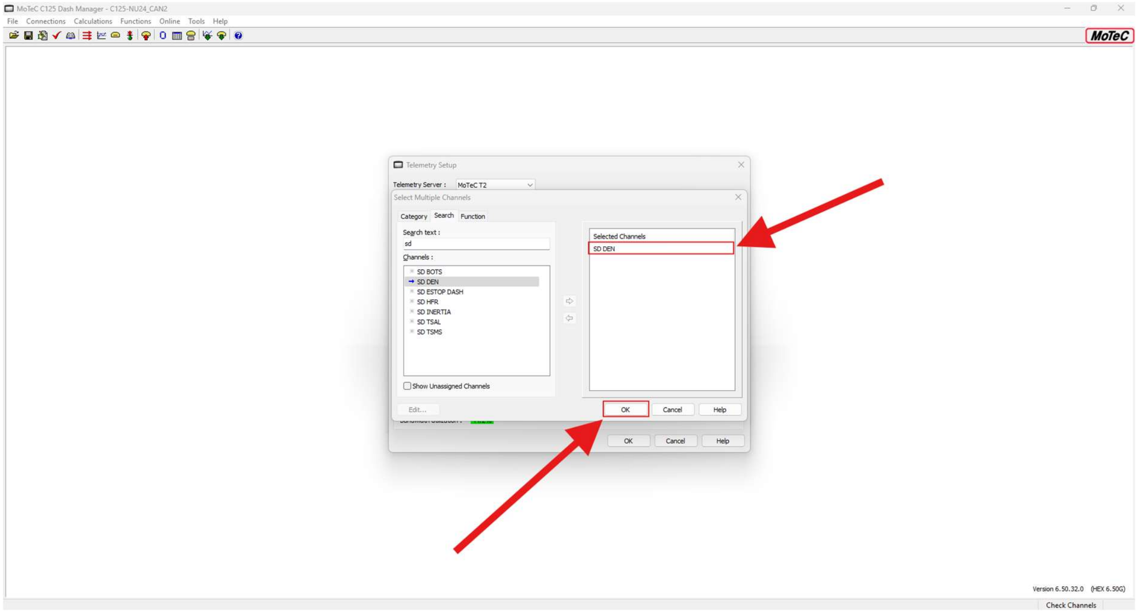
- Ensure MoTeC T2 is selected in the 'Telemetry Server' option
- Ensure the Real Time tab is selected
- Click 'Add…'



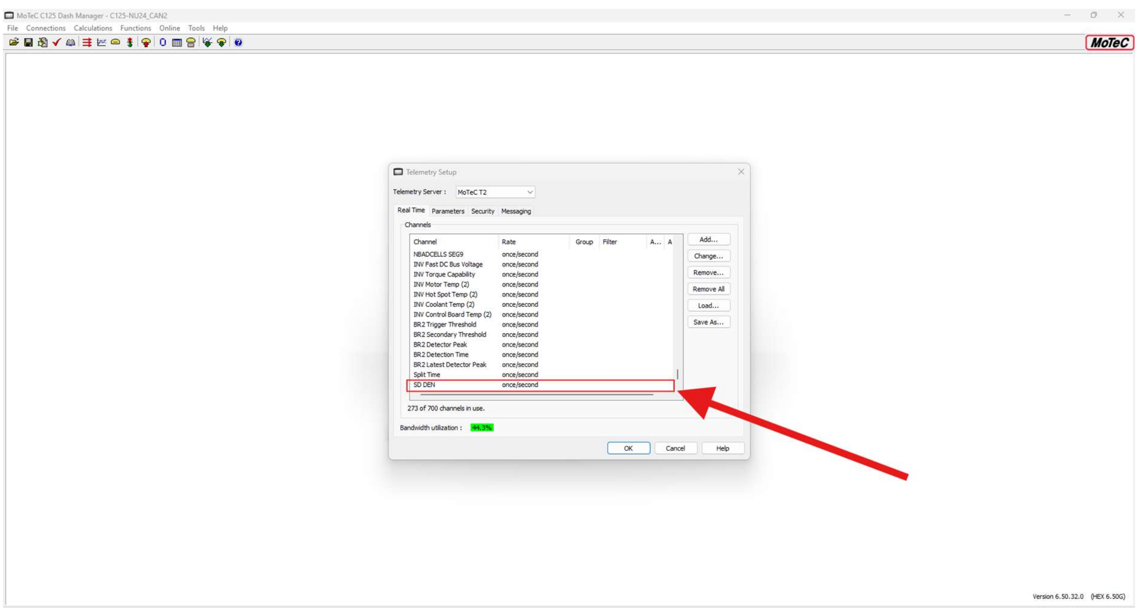
- Find the desired message(s) to add



- Ensure the message(s) are in the 'Selected Channels' box, click 'Ok'



- Scroll to check that the new message(s) has been added correctly
    - In this section, it is okay to leave it as once/second here, this is how often the signal will update on i2 when on track side

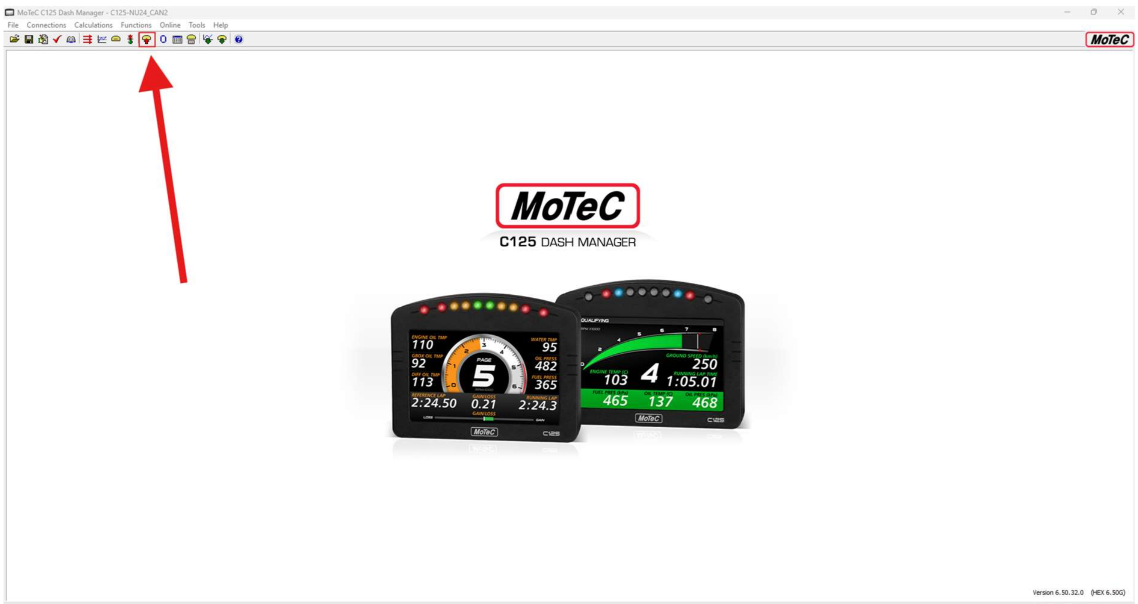


- Select 'Ok', then done!
- Then make sure to click 'Save File' in the top left-hand corner
    - Dash Manager is a funky program, and if this isn't done then changes won't update through the whole program correctly

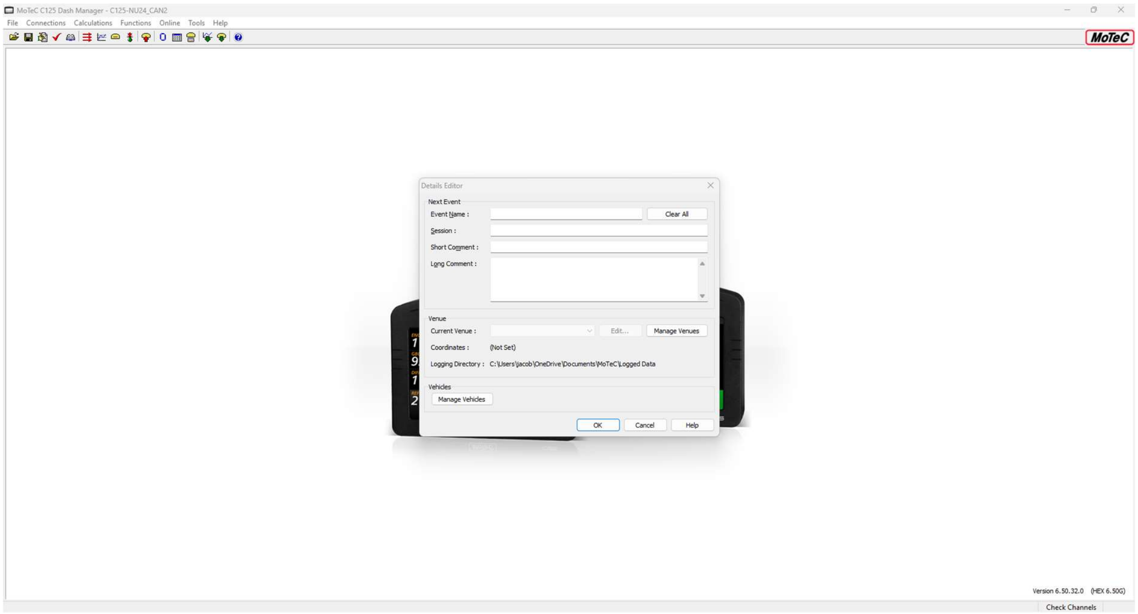
- Next, upload the config to the Dash. See the Send Config section of this document

Send Configuration to C125

- Ensure that the desired config file is loaded (see the Opening an Existing Configuration File section for more information)
- Ensure the C125 Dash is connected through an ethernet cable to the computer uploading the config
    - On NU23, NU24 and NU25, this port is located on the right-hand side of the Dash Electronic Node (DEN)
    - This connection can be very dodgy, if the cable is connected and it is not connecting, try a different port, reconnect it, wiggle it around a bit, it should connect after a bit of messing about
- Click 'Send Configuration'



- After a successful connection, the following window should pop up
    - Fill these out accordingly (on the Chungus, these fields should already be prefilled with whatever was previously entered)
    - See the Clearing Data Logger at the start of the Day section for more information



- After filling out the fields and selecting ‘Ok’, the configuration will start uploading. This will take some time to upload so go treat yourself to a snack from TA Snackies (if it hasn’t been shut down by the ATO)
- Then make sure to click ‘Save File’ in the top left-hand corner
    - Dash Manager is a funky program, and if this isn’t done then changes won’t update through the whole program correctly

- To see if the new message(s) has been configured correctly, set up the MoTeC transceiver box and Chungus with T2 and i2 set up accordingly (see the Setting up MoTeC for a Track Day section for more information), add the message(s) to the i2 worksheet (see the Editing Worksheets (channels and groups) section for more information), and turn on LV! The message(s) should now be updating! If they are not, try going back through the steps and double check that a step wasn't missed. The author recommends to not use ChatGPT for this stuff (at least at the time of writing, July 2025, as it is not good at knowing what to do, and often gives false information)

Get Configuration from C125

- Ensure the C125 Dash is connected through an ethernet cable to the computer uploading the config
  - On NU23, NU24 and NU25, this port is located on the right-hand side of the Dash Electronic Node (DEN)
  - This connection can be very dodgy, if the cable is connected and it is not connecting, try a different port, reconnect it, wiggle it around a bit, it should connect after a bit of messing about
- Click 'Get Configuration'

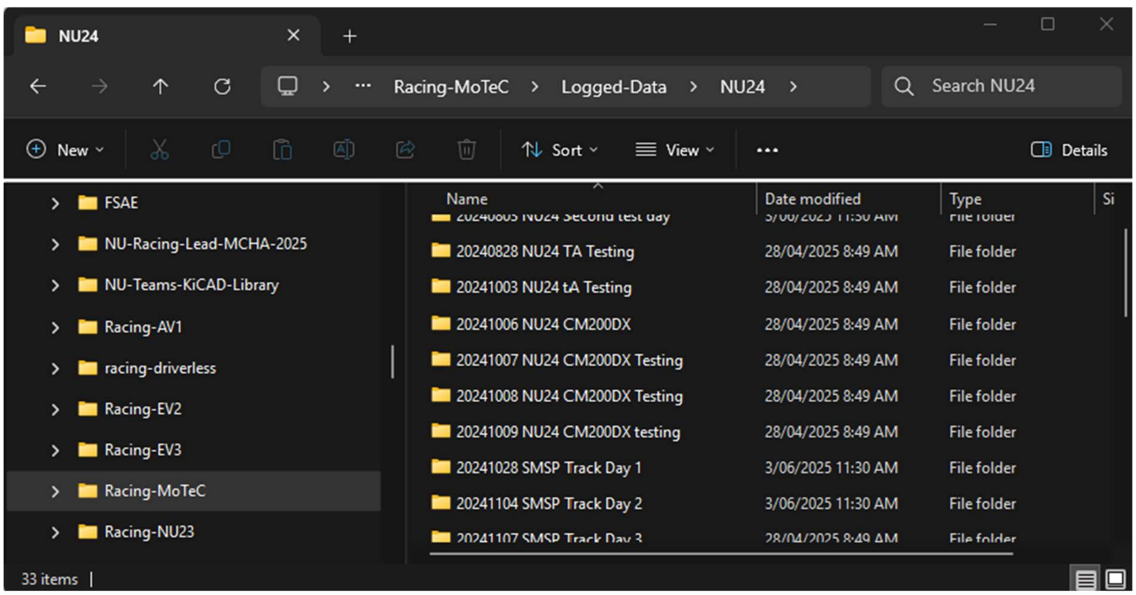
- It should connect, and the configuration should be able to be edited and changed

Pulling Data

- Ensure the C125 Dash is connected through an ethernet cable to the computer uploading the config
    - On NU23, NU24 and NU25, this port is located on the right-hand side of the Dash Electronic Node (DEN)
- Select ‘Get Logged Data’
    - This connection can be very dodgy, if the cable is connected and it is not connecting, try a different port, reconnect it, wiggle it around a bit, it should connect after a bit of messing about

- After a successful connection, the following window should pop up
    - Fill these out accordingly (on the Chungus, these fields should already be prefilled with whatever was previously entered)
    - Ensure the correct log file has been selected in the ‘Current Venue’ section. If this is not configured correctly, then after pulling data, the file will be stored in a random location and impossible to find
        - Log files are classically stored in the ‘Logged-Data’ folder in the Racing-MoTeC Github Repo, in the relevant car name folder, in a folder with the following naming scheme:
            - YYYYMMDD name-of-test (example given below)
        - For this step, the Racing-MoTeC repo is required, see the Racing-MoTeC Repository section for more information
    - See the Clearing Data Logger at the start of the Day section for more information


MoTeC
Details Editor
Next Event
Event Name :
Clear All
Session :
Short Comment :
Long Comment :
Venue
Current Venue :
Edit...
Manage Venues
Coordinates :
(Not Set)
Logging Directory :
C:\Users\jacob\OneDrive\Documents\MoTeC\Logged Data
Vehicles
Manage Vehicles
OK
Cancel
Help
Check Channels


NU24
Racing-MoTeC
Logged-Data
NU24
Search NU24
New
Sort
View
Details
FSAE
NU-Racing-Lead-MCHA-2025
NU-Teams-KiCAD-Library
Racing-AV1
racing-driverless
Racing-EV2
Racing-EV3
Racing-MoTeC
Racing-NU23
Name
Date modified
Type
20240828 NU24 TA Testing
28/04/2025 8:49 AM
File folder
20241003 NU24 tA Testing
28/04/2025 8:49 AM
File folder
20241006 NU24 CM200DX
28/04/2025 8:49 AM
File folder
20241007 NU24 CM200DX Testing
28/04/2025 8:49 AM
File folder
20241008 NU24 CM200DX Testing
28/04/2025 8:49 AM
File folder
20241009 NU24 CM200DX testing
28/04/2025 8:49 AM
File folder
20241028 SMSP Track Day 1
3/06/2025 11:30 AM
File folder
20241104 SMSP Track Day 2
3/06/2025 11:30 AM
File folder
33 items

Live Signal Monitoring using Dash Manager

- Ensure the C125 Dash is connected through an ethernet cable to the computer uploading the config
    - On NU23, NU24 and NU25, this port is located on the right-hand side of the Dash Electronic Node (DEN)
    - This connection can be very dodgy, if the cable is connected and it is not connecting, try a different port, reconnect it, wiggle it around a bit, it should connect after a bit of messing about
- Select Monitor Channels

- Once it connects, a full list of all the messages that the MoTeC transmits should be shown, bit annoying to scroll through and find the message(s) that are wanting to be monitored, but such is life

i2

Live Signal Monitoring using i2

- Data tab (on right hand screen) > 'Telemetry' > 'Add…' (bottom left) > select name of telemetry source (this should show up if the motec transceiver box and T2 is opened and the graphs are green) > Ok > let it connect > Close > data should be updating now

Opening a log file

Editing Worksheets (channels and groups)

Display Creator

To edit the screen with Display Creator, a DBC-style file must be made from within Dash Manager, exported to Display Creator, then a Display Creator Package is exported back to Dash Manager to be uploaded to the C125. After doing this, the desired signals will be displayed on the screen.

WARNING: The MoTeC C125 has been known to have issues with being overloaded with having too many signals being displayed. This can be avoided by only uploading the minimum number of signals to the display, and nothing more. The issues that the C125 has include signal values not displaying correctly and the button to change pages not working.

> NOTE: only the signals that are added to be displayed on the dash count within this requirement. The display does not care about all the other signals that are on the CAN Bus

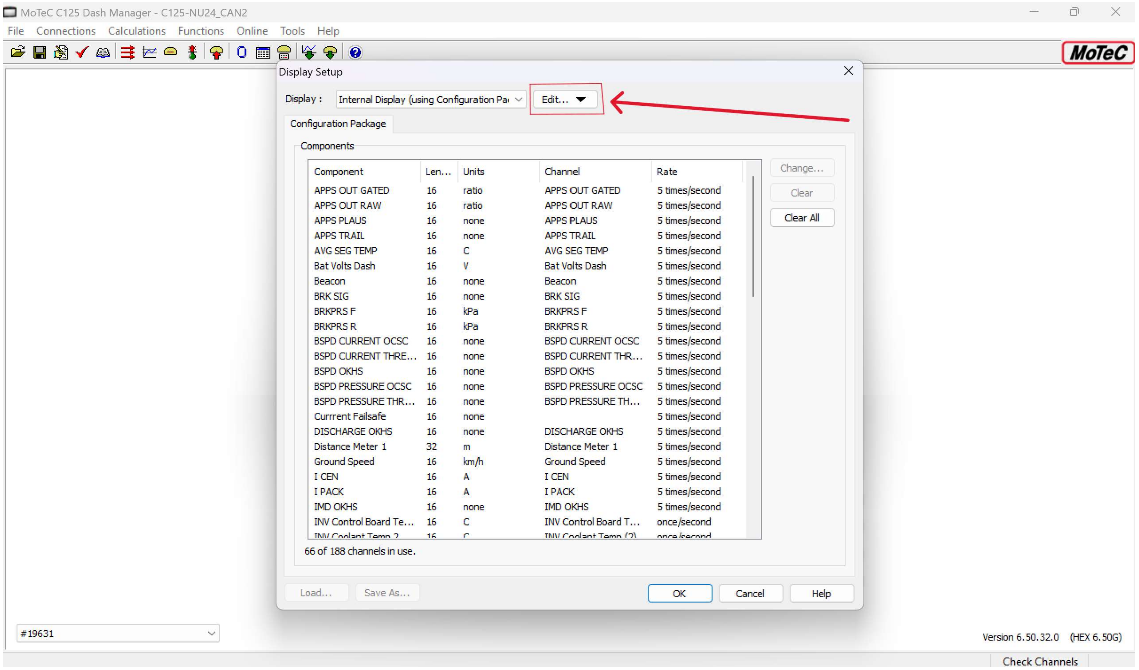
The following steps detail how to:

- export the DBC-style file from Dash Manager to Display Creator
    - -> Exporting DBC-Style File from Dash Manager to Display Creator
- assign signals to values on the display
    - -> Assigning Signals to Values on Display
- export the Display Creator Package back to Dash Manager
    - -> Exporting Display Creator Package to Dash Manager
- upload to the C125.
    - -> Uploading Display Creator Package to C125 from Dash Manager

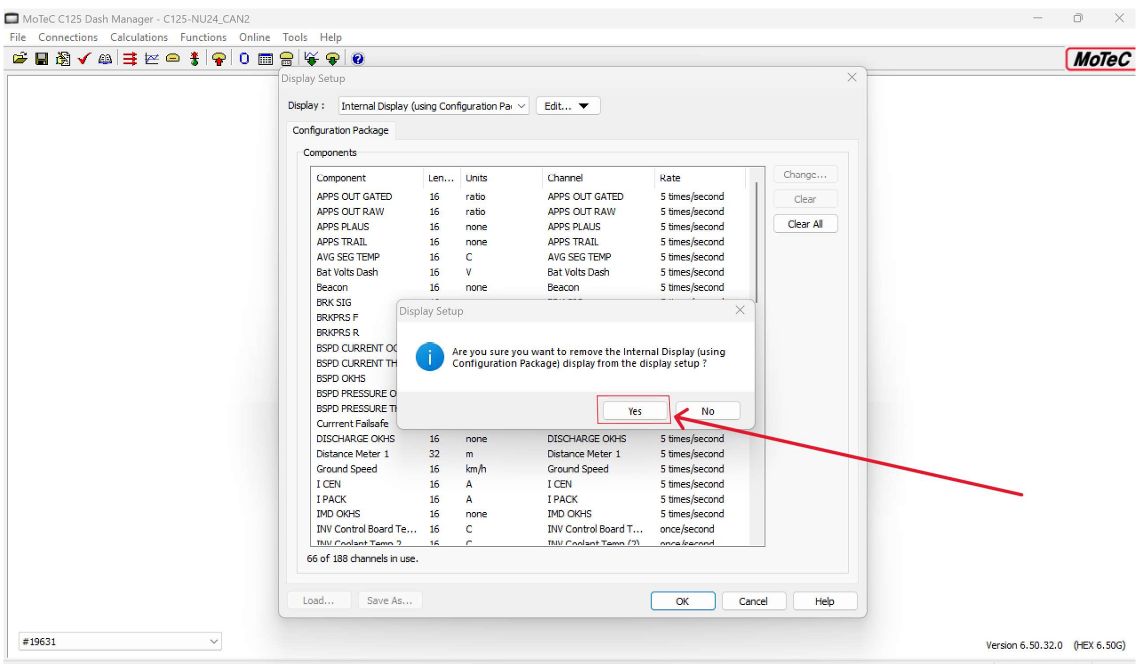
Exporting DBC-Style File from Dash Manager to Display Creator

- Open Dash Manger, open the Dash Configuration File (see Opening an Existing Configuration File)
- Navigate to Functions > Display…

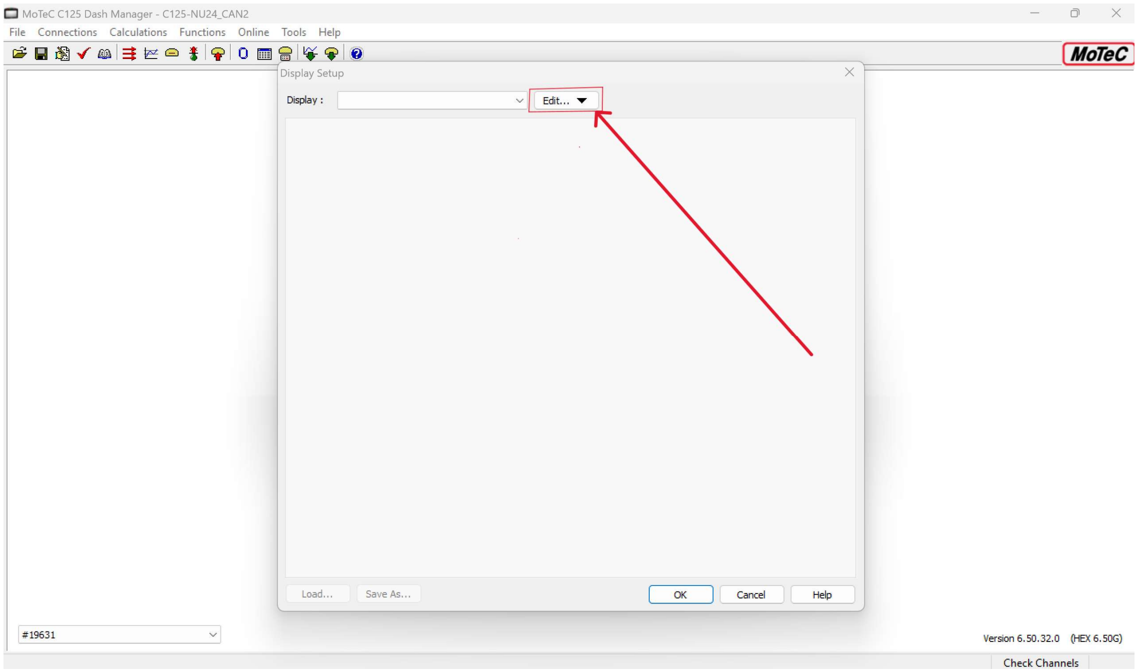
- If there are already channels (signals) added to the Configuration Package, they must be removed to start fresh. To do this, select Edit… > Remove



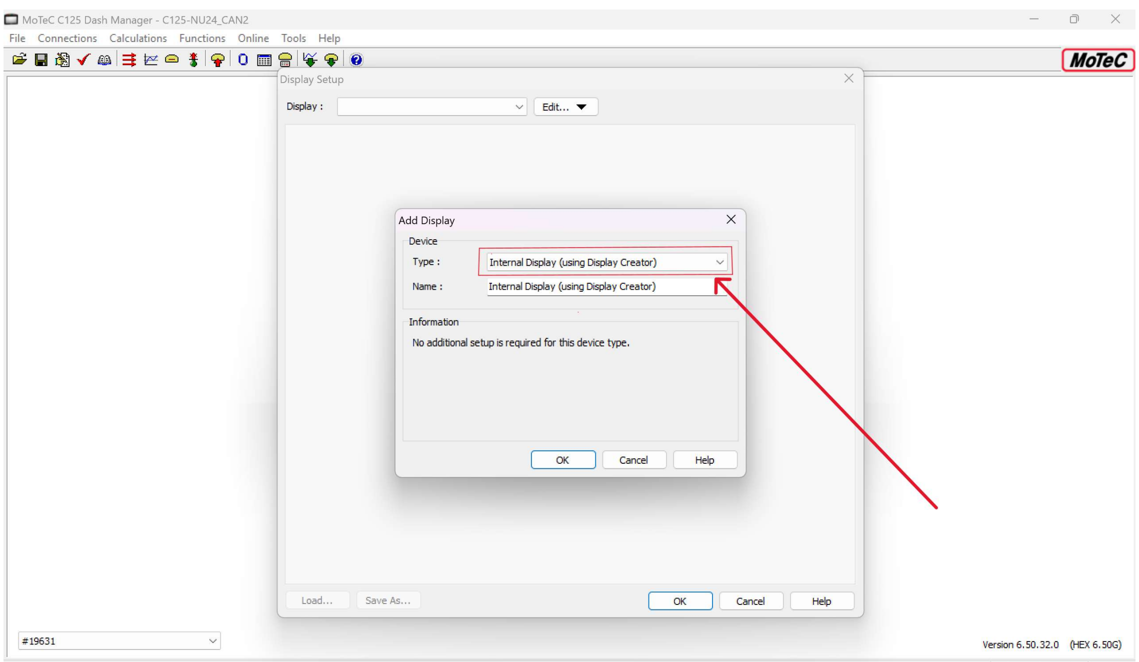
- Select Yes



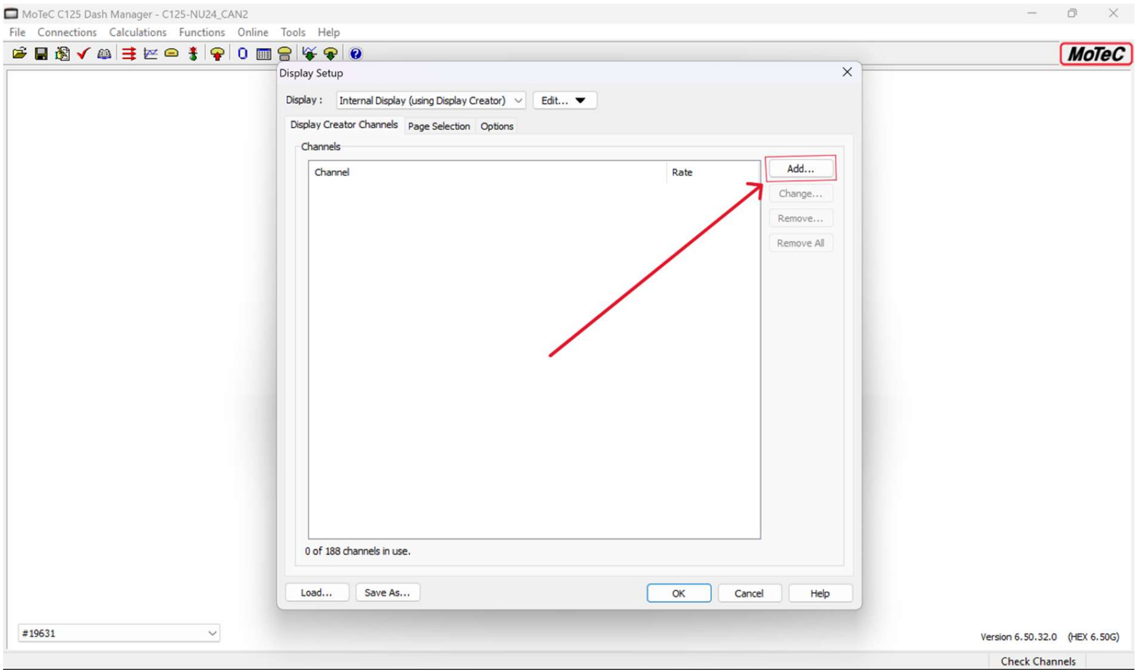
- Now with a blank display, select Edit… again > Add



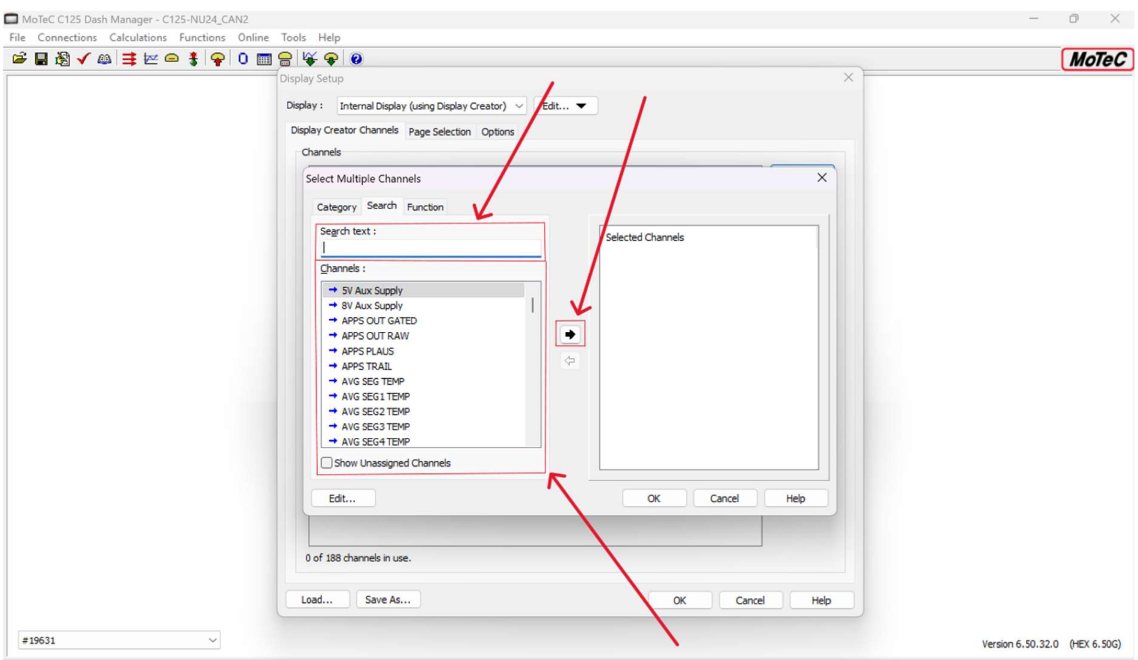
- Select the drop-down box, change ‘Internal Display (Fixed Layout)’ to ‘Internal Display (Using Display Creator)’, then select ‘Ok’



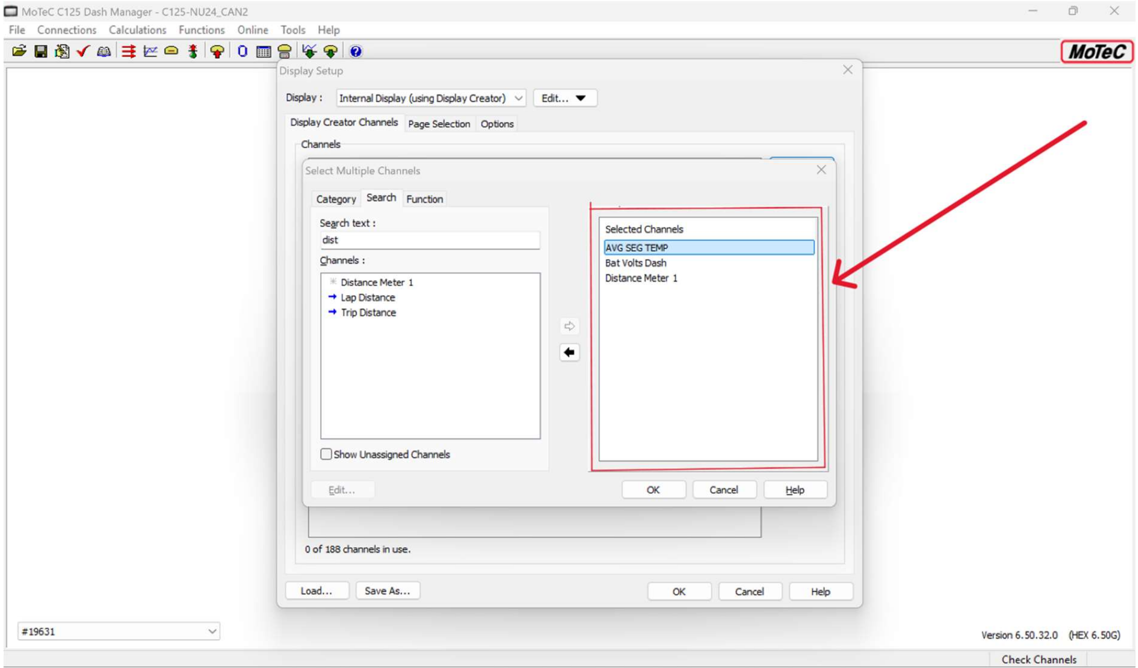
- Now select Add…



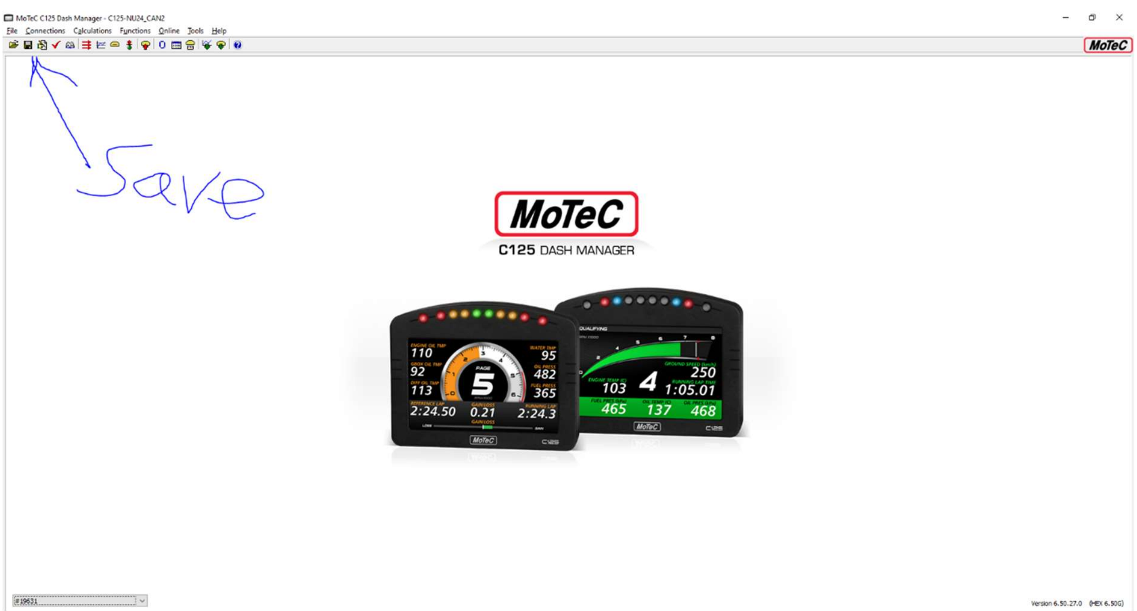
- Then, the signals to be displayed on the dash can be selected, search for signals with the 'Search Text:' box, or, scroll through the list shown. When a signal is selected, the arrow in the middle must be clicked to bring the signal to the 'Selected Channels' Section
    - NOTE: multiple signals can be selected before clicking the arrow to bring them across



- Signals added to the ‘Selected Channels’:



- Click ‘Ok’ to close the Select Multiple Channels window, then ‘Ok’ to close the Display Setup window. Then click save!



- Then reopen Display Setup (Functions > Display…)

Assigning Signals to Values on Display

Exporting Display Creator Package to Dash Manager

Uploading Display Creator Package to C125 from Dash Manager

Pages

DBC

Other Functions

Settings, alarms, video clips, simulate

T2

- T2 can only be opened when the 'MoTeC Discovery' task is not running
    - MoTeC Discovery task opens when MoTeC Dash Manager is running and connected to the motec -> happened with matt on the boat

Using T2 with i2 Pro

Set up on the Car

GPS

Transceiver

Dash

Ethernet Access Point